Lecture Notes on Effective Field Theories in Cosmology

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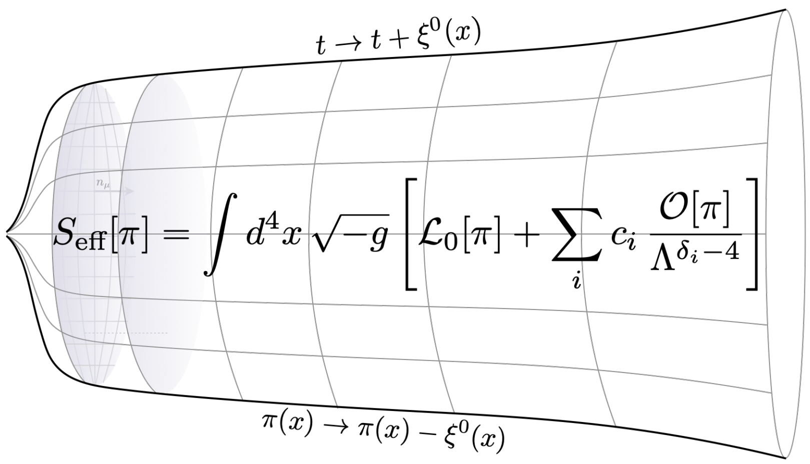

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Abstract

In these notes I review the construction of the effective field theory (EFT) of cosmological perturbations around a FLRW background. I mainly focus on the geometric construction of the EFT of inflation, but the same logic and structure is straightforwardly extended to the EFT of dark energy. The construction is based on the fact that cosmological perturbations are nothing but Goldstone bosons of the *spontaneously* broken time diffeomorphism invariance of FLRW backgrounds. By writing the action in unitary gauge we capture the gravitational dynamics along a preferred spacetime foliation, by applying the Stückelberg mechanism we restore the full diffeomorphism invariance of the theory and the Goldstone appears explicit in the action, then by taking the decoupling limit we end up with the effective action describing fluctuations around FLRW.

Contents

Disclaimer: These notes are still under construction!

Why Effective Field Theory in Cosmology?

Before explaining EFTs, let's begin by what we do in cosmology?

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$$

Solutions:

- Black Holes
- Gravitational Waves
- Expanding Universes

$$ds^2 = -dt^2 + a^2(t)\delta_{ij}dx^i dx^j$$

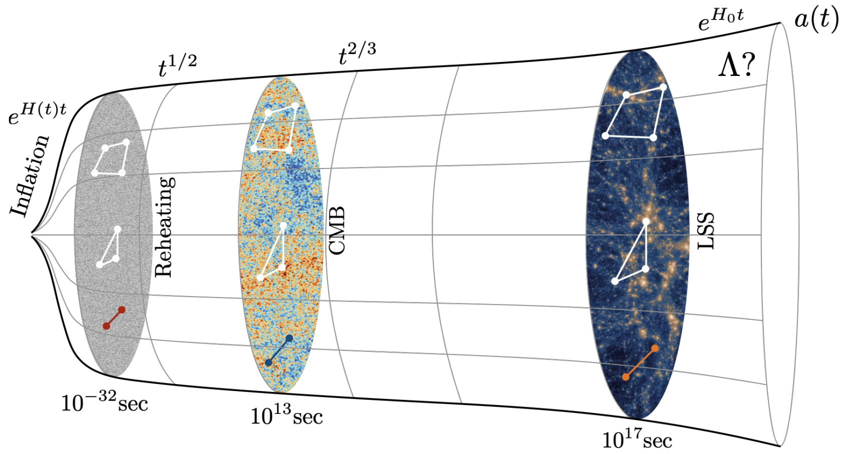

Figure 1: The correlations in late-time cosmological observables—like the distribution of the large-scale structure (LSS) or the anisotropies of the cosmic microwave background (CMB)—can be traced back in time to primordial correlations on the "reheating surface" (marking the initial surface of the hot Big Bang universe or the final surface at the end of inflation). Imprinted in these correlations is the physics of the inflationary era.

The Principles of EFT

An EFT as the name suggest, is an effective, low-energy (IR) description of the relevant degrees of freedom involved in a physical phenomenon at a given energy scale.

Top-bottom: The UV theory is *known*

$$\mathcal{L}_{\rm UV}[\ell, H] = \mathcal{L}_\ell[\ell, m] + \mathcal{L}_H[H, M] + \mathcal{L}_{\rm int}[\ell, H]$$

If $m \ll M$ we can integrate out the heavy field H and build and effective action for the light degree of freedom

$$e^{iS_{\rm eff}[\ell]} = \int \mathcal{D}H\, e^{iS_{\rm UV}[\ell, H]},$$

and expand the action in local operators built from ℓ

$$\mathcal{L}_{\rm eff}[\ell] = \mathcal{L}_\ell^{\rm ren}[\ell] + \sum_i \frac{C_i(\mu)}{M^{d_i-4}}\,\mathcal{O}_i[\ell]$$

Bottom-up: The UV theory is *unknown*.

- Identify the relevant degrees of freedom involved in the physical phenomenon Φ_i.
- Identify the energy scale of the problem Λ.
- Identify the (broken) symmetries.
- Write all the operators $\mathcal{O}$ compatible with the symmetries.
- Compute observables, do and experiment and fix the Wilson coefficients C_n.

$$\mathcal{L}_{\rm EFT}[\Phi_i] = \mathcal{L}_0[\Phi_i] + \sum_n \frac{C_n}{\Lambda^{d_i-4}}\,\mathcal{O}_n[\Phi_i]$$

The Wilson coefficients encode the information about the UV physics. Example: Non-linear sigma model of Nucleons and Pions is the low-energy EFT of QCD.

These three lectures build a single object and use it twice. A cosmological spontaneously breaks time diffeomorphisms while leaving spatial diffeomorphisms intact. This breaking pattern selects a preferred foliation of spacetime. The geometric quantities covariant under the residual (spatial) symmetry — the ADM/foliation variables — organize the most general action of the fluctuations. The *same* action describes two physical regimes:

- **Inflation:** quasi–de Sitter background, matter irrelevant, the Goldstone π is essentially the curvature perturbation ζ. We extract the sound speed, the dispersion relation, the power spectrum, and non-Gaussianities.
- **Dark energy:** general FLRW background with $w(t) \neq -1$, matter present, gravity kept dynamical. The operators that were decoration during inflation become the physics.

The throughline to sell relentlessly: *the zoo of models is a map of points in a coefficient space, and the same broken-clock construction covers inflation and late-time acceleration.*

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1 Spontaneous Symmetry Breaking in Flat Space

Goal: end with the geometric building blocks identified. No action yet.

1.1 Counting degrees of freedom

1.1.1 Massless spin − 1

Let recall that a massless spin-1 particle carries only the two transverse helicities $\lambda = \pm 1$. No longitudinal state: there is no rest frame to define one.

$$\mathcal{L}[A_\mu] = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad \partial^\mu F_{\mu\nu} = \Box A_\nu - \partial_\nu(\partial^\mu A_\mu) = 0. \tag{1}$$

Covariant count — gauge symmetry. Insert a plane wave $A_\mu = \epsilon_\mu \, e^{ik\cdot x}$ with polarization ϵ_μ (four components). Because the free theory is linear and translation invariant, every solution is a superposition of such modes, and counting polarizations mode by mode *is* counting the physical content. The field equation gives

$$k^2\epsilon_\nu - k_\nu\,(k\cdot\epsilon) = 0. \tag{$\star$}$$

On the light cone, $k^2 = 0$ this forces

$$k\cdot\epsilon = 0 \qquad \text{(transversality)}: \quad 4 \to 3. \tag{2}$$

Now use the gauge symmetry. Under $A_\mu \to A_\mu + \partial_\mu\alpha$ with $\alpha = ice^{ik\cdot x}$,

$$\epsilon_\mu \to \epsilon_\mu + ck_\mu, \tag{3}$$

and this shift preserves $k\cdot\epsilon = 0$ because $k^2 = 0$. Polarizations differing by a multiple of k_μ are therefore *the same physical state*:

$$3 \to 2 \quad \text{physical polarizations} \tag{4}$$

The two deletions to keep straight: transversality is one condition; the residual gauge identication $\epsilon \sim \epsilon + ck$ is the second. Together they take $4 \to 2$.

1.1.2 Massive spin − 1

Add a mass by hand:

$$\mathcal{L}[A_\mu; m] = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}m^2 A_\mu A^\mu, \qquad \partial^\mu F_{\mu\nu} - m^2 A_\nu = 0. \tag{5}$$

The mass term breaks gauge invariance: under $A_\mu \to A_\mu + \partial_\mu\alpha$ it shifts by $m^2 A_\mu \partial^\mu\alpha + \cdots \neq 0$. So there is *no gauge freedom to spend.* Take the divergence ∂^ν of the field equation. The first term dies identically ($\partial^\nu\partial^\mu F_{\mu\nu} = 0$ by antisymmetry), leaving

$$m^2\,\partial^\nu A_\nu = 0 \quad \Longrightarrow \quad \partial^\mu A_\mu = 0. \tag{6}$$

Transversality reappears — but now as a *consequence of the dynamics*, not a choice. Feeding $\partial \cdot A = 0$ back into the field equation reduces it to

$$(\Box - m^2)A_\nu = 0 \implies k^2 = -m^2, \qquad \text{i.e.} \quad \omega^2 = \vec{k}^2 + m^2. \tag{7}$$

The dispersion relation is again an output. And because the mode is massive, we may boost to the rest frame $k^\mu = (m, 0, 0, 0)$, $k_\mu = (-m, 0, 0, 0)$, and count in components:

$$k \cdot \epsilon = -m\,\epsilon^0 = 0 \implies \epsilon^0 = 0, \qquad \epsilon^\mu = (0,\ \epsilon^1,\ \epsilon^2,\ \epsilon^3). \quad \boxed{\text{3 physical: } \lambda = -1, 0, +1.} \tag{8}$$

Step 1 (transversality kills ϵ^0) runs exactly as in Maxwell. But there is *no Step 2*: with no gauge symmetry, $\epsilon \to \epsilon + c\,k$ is not a redundancy and deletes nothing. The three spatial polarizations all survive, the new one being the longitudinal $\epsilon^\mu = (0, 0, 0, 1)$.

The longitudinal mode in motion. The three states above live in the rest frame. Nothing forces the particle to be at rest: boost along z to $k^\mu = (E, 0, 0, k)$, $E^2 = k^2 + m^2$. Polarizations are vectors, so they ride the same Lorentz boost Λ (in the t–z plane) that carried $(m, 0, 0, 0) \to (E, 0, 0, k)$:

$$\Lambda^\mu{}_\nu = \begin{pmatrix} E/m & 0 & 0 & k/m \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ k/m & 0 & 0 & E/m \end{pmatrix}. \tag{9}$$

The transverse states $\epsilon_{(1)}, \epsilon_{(2)}$ lie in x, y and are untouched. The longitudinal one is boosted:

$$\epsilon^\mu_L = \Lambda^\mu{}_\nu\, \epsilon^\nu_{(3)} = \Lambda^\mu{}_\nu\, (0, 0, 0, 1)^\nu = \frac{1}{m}\,(k, 0, 0, E). \tag{10}$$

Checks that it is still a legitimate polarization: $k \cdot \epsilon_L = \frac{1}{m}(-Ek + kE) = 0$ and $\epsilon_L \cdot \epsilon_L = \frac{1}{m^2}(-k^2 + E^2) = 1$. *This is not a fourth polarization, and not a combination of the rest-frame basis* — that basis all has $\epsilon^0 = 0$, whereas $\epsilon^0_L = k/m \neq 0$. The reason: the physical polarizations span the 3-plane orthogonal to k^μ, and boosting tilts k^μ, so the plane tilts with it. In terms of fixed lab vectors,

$$\epsilon_L = \frac{E}{m}\,\epsilon_{(3)} + \frac{k}{m}\,\hat{t}, \qquad \hat{t} \equiv (1, 0, 0, 0), \tag{11}$$

i.e. the longitudinal state *mixes with the timelike direction* $\hat{t}$ — exactly the unphysical mode transversality removed at rest. That admixture, $\frac{k}{m}\hat{t}$, is the piece that grows as E/m.

The same mode, two fates. At high energy

$$\epsilon^\mu_L = \frac{1}{m}(k, 0, 0, E) = \frac{k^\mu}{m} + \mathcal{O}\Big(\frac{m}{E}\Big), \tag{12}$$

since $\frac{1}{m}(E, 0, 0, k) = k^\mu/m$ and the remainder $\frac{1}{m}(k - E, 0, 0, E - k)$ is $\mathcal{O}(m/E)$ using $E - k \simeq m^2/2E$. The leading direction k^μ/m is precisely the temporal–longitudinal combination that was *pure gauge* (unphysical) in Maxwell: the would-be gauge mode of the massless theory is the physical longitudinal mode of the massive one. Its growth with energy is the whole UV story of the massive vector, and Section 4 shows the Goldstone captures it.

One line worth saying. For abelian $U(1)$ this by-hand mass is perfectly healthy: the theory is unitary at all energies. For a non-abelian vector it is not — massive Yang–Mills violates unitarity at $E \sim m/g$ and *requires* a Higgs completion. That is why the Higgs mechanism is mandatory in the electroweak theory, and it is the thread we pick up in Section 6.

1.2 Stückelberg mechanism and the decoupling limit

To see the third polarization *as* a scalar, introduce a field π by

$$A_\mu \longrightarrow A_\mu + \frac{1}{m}\partial_\mu \pi, \tag{13}$$

promoting π to a genuine field (not a gauge parameter). The Lagrangian becomes

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}m^2 A_\mu A^\mu - m\, A_\mu \partial^\mu \pi - \frac{1}{2}\,\partial_\mu \pi\, \partial^\mu \pi, \tag{14}$$

now invariant under a *restored* gauge symmetry

$$A_\mu \to A_\mu + \partial_\mu \alpha, \qquad \pi \to \pi - m\,\alpha. \tag{15}$$

We added one field and one gauge symmetry together, so the count is untouched: 2 (gauge A_μ) $+1$ (π) $= 3$. Stückelberg exposes the third polarization of the massive photon as a scalar — the would-be Goldstone.

The name, abelian preview. The restored gauge symmetry can be spent to set $\pi = 0$. That choice is called *unitary gauge*, and it simply returns the Proca Lagrangian of Section 2. So Proca *is* the abelian theory in unitary gauge; Stückelberg is the act of undoing that gauge to expose the Goldstone. The name earns its keep in the non-abelian case (Section 7), where there are several Goldstones to remove at once.

1.3 The origin of the mass

1.4 Why EFT in cosmology

Nature separates into scales, and we treat one at a time. An EFT parametrizes our ignorance of the short-distance physics: fix the light degrees of freedom and the symmetries, write every operator the symmetries allow, and let observations fix the coefficients. "Everything not forbidden is compulsory."

Frame gravity itself this way, since the audience already owns it:

$$S_{\rm grav} = \int \mathrm{d}^4x\,\sqrt{-g}\left[\frac{M_{\rm pl}^2}{2}R + c_1 R^2 + c_2 R_{\mu\nu}R^{\mu\nu} + \frac{1}{\Lambda^2}\left(d_1 R^3 + \cdots\right)\right], \qquad \Lambda \lesssim M_{\rm pl}. \tag{16}$$

General relativity is the leading term of a tower. That reframing — "you already know an EFT" — is the entry point.

1.5 Toy model I — global $U(1)$

Start with the cleanest symmetry breaking, to isolate the Goldstone logic before any gravity:

$$\mathcal{L} = -\partial_\mu \phi^\dagger \partial^\mu \phi + \mu^2 \phi^\dagger \phi - \tfrac{1}{4}\lambda(\phi^\dagger\phi)^2, \qquad \phi \to e^{i\beta}\phi. \tag{17}$$

For $\mu^2 > 0$ the $U(1)$ breaks. Write $\phi = \frac{1}{\sqrt{2}}(v+\rho)\,e^{i\pi}$, with v the location of the potential minimum (fixed below). The phase π is a massless Goldstone; ρ is heavy. Integrating out ρ leaves a derivative expansion for π, worked out explicitly here:

Integrating out ρ : On $\phi = \frac{1}{\sqrt{2}}Re^{i\pi}$, $R \equiv v + \rho$, the potential $V = -\mu^2\phi^\dagger\phi + \frac{1}{4}\lambda(\phi^\dagger\phi)^2$ becomes

$$V(R) = -\tfrac{1}{2}\mu^2 R^2 + \tfrac{1}{16}\lambda R^4. \tag{18}$$

No linear term in ρ (tadpole cancellation — v genuinely extremizes V) fixes v:

$$V'(v) = 0 \implies -\mu^2 v + \tfrac{1}{4}\lambda v^3 = 0 \implies v^2 = \frac{4\mu^2}{\lambda}, \tag{19}$$

and the quadratic term gives the ρ mass,

$$m_\rho^2 \equiv V''(v) = -\mu^2 + \tfrac{3}{4}\lambda v^2 = 2\mu^2. \tag{20}$$

Expanding $\mathcal{L}$ to quadratic order in ρ around this minimum,

$$\mathcal{L} \supset -\tfrac{1}{2}(\partial\rho)^2 - v\rho(\partial\pi)^2 - \tfrac{1}{2}m_\rho^2\rho^2, \tag{21}$$

the heavy field's equation of motion is $(\Box - m_\rho^2)\rho = v(\partial\pi)^2$. Solve it order by order in E/m_ρ: at leading order drop $\Box\rho$ (each extra power of $\Box/m_\rho^2$ costs two more derivatives on π), giving the algebraic solution

$$\rho_\star = -\frac{v}{m_\rho^2}(\partial\pi)^2 + \mathcal{O}\left(\frac{\partial^2}{m_\rho^4}\right). \tag{22}$$

Substituting $\rho = \rho_\star$ back into $\mathcal{L}$ — valid at tree level, since $\rho_\star$ extremizes the ρ-dependent terms, so the correction is second order in $(\rho - \rho_\star)$ — leaves

$$\mathcal{L}_\pi = -\tfrac{1}{2}v^2(\partial\pi)^2 + \frac{v^2}{2m_\rho^2}(\partial\pi)^4 + \cdots . \tag{23}$$

In canonical variables $\pi_c \equiv v\pi$ this is exactly

$$\mathcal{L}_\pi = -\tfrac{1}{2}(\partial_\mu\pi_c)^2 + c_1\frac{(\partial_\mu\pi_c)^4}{v^4} + \cdots , \qquad c_1 = \frac{v^2}{2m_\rho^2} = \frac{1}{\lambda}, \tag{24}$$

using $m_\rho^2 = \frac{1}{2}\lambda v^2$ from above.

1.6 Toy model II — gauge $U(1)$, Stückelberg, decoupling

Now gauge the symmetry, $D_\mu = \partial_\mu + igA_\mu$. In unitary gauge ($\pi \equiv 0$) the vector eats the Goldstone and becomes massive,

$$\mathcal{L} = -\tfrac{1}{4}F_{\mu\nu}^2 - \tfrac{1}{2}m^2A_\mu^2 + \cdots , \qquad m^2 = g^2v^2. \tag{25}$$

To probe high energies, *undo* the gauge fixing — the Stückelberg trick:

$$A_\mu \to A_\mu + \frac{\partial_\mu\pi}{g}, \qquad \mathcal{L}_2 = -\tfrac{1}{4}F_{\mu\nu}^2 - \tfrac{1}{2}(\partial_\mu\pi_c)^2 - \tfrac{1}{2}m^2A_\mu^2 + m\,\partial_\mu\pi_c A^\mu, \tag{26}$$

with $\pi_c \equiv f_\pi\pi$, $f_\pi \equiv m/g$. The mixing $m\,\partial_\mu\pi_c A^\mu$ carries one fewer derivative than the kinetic term, so it fades at high energy. Take the decoupling limit,

$$g \to 0, \quad m \to 0, \quad f_\pi = m/g \text{ fixed.} \tag{27}$$

Above $E_{\text{mix}} = m$, the scattering of the longitudinal vector equals the scattering of the Goldstone (Goldstone equivalence theorem).

1.7 Breaking time diffeomorphisms

Here is the conceptual heart. A homogeneous rolling field $\bar\phi(t)$ — the clock — picks out a time direction and breaks time diffeomorphisms. Under a time shift the clock moves, so the fluctuation is a Goldstone:

$$t \to t + \pi(x), \qquad \delta\phi(t,\mathbf{x}) = \bar\phi(t+\pi) - \bar\phi(t) \simeq \dot{\bar\phi}\,\pi. \tag{28}$$

Unitary gauge uses the time-diff freedom to sit on the slice where the clock is uniform ($\delta\phi \equiv 0$): the Goldstone is eaten by the metric, exactly as in the gauge toy model. What survives:

$$\text{Diff} \longrightarrow \text{Diff}_{\text{spatial}} \qquad \Longrightarrow \qquad \text{a preferred spacetime foliation.} \tag{29}$$

1.8 ADM as the language of the foliation, the building blocks

Spacetime decomposition $\mathcal{M}_4 \simeq \Sigma_3 \times \mathbb{R}$

A preferred foliation demands the ADM decomposition:

$$\mathrm{d}s^2 = -N^2\,\mathrm{d}t^2 + h_{ij}\left(\mathrm{d}x^i + N^i\mathrm{d}t\right)\left(\mathrm{d}x^j + N^j\mathrm{d}t\right), \tag{30}$$

i.e., reading off components,

$$g_{00} = -N^2 + N_iN^i, \qquad g_{0i} = N_i, \qquad g_{ij} = h_{ij}, \qquad N_i \equiv h_{ij}N^j. \tag{31}$$

Invert this block matrix (impose $g^{\mu\nu}g_{\nu\lambda} = \delta^\mu_\lambda$ and check the g^{0i} and g^{ij} blocks close on themselves) to get

$$g^{00} = -\frac{1}{N^2}, \qquad g^{0i} = \frac{N^i}{N^2}, \qquad g^{ij} = h^{ij} - \frac{N^iN^j}{N^2}, \tag{32}$$

with h^{ij} the inverse of h_{ij} (not g_{ij}'s naive inverse). Note N and N^i enter $g_{\mu\nu}$ without time derivatives: their own equations of motion are constraints (Hamiltonian and momentum), not propagating fields — used freely below without re-deriving it.

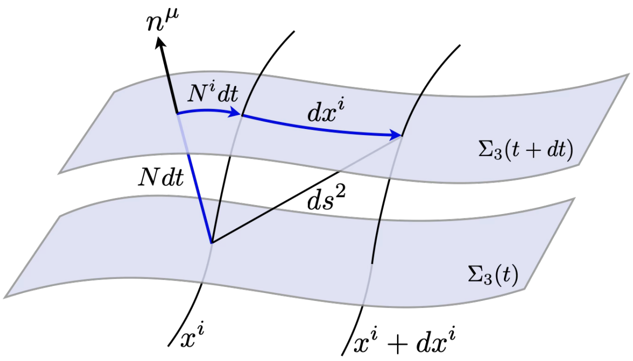

Figure 2: The ADM decomposition of the metric. The blue arrows are the components of the metric, namely, lapse of time Ndt, a shift of hypersurface due to the time flow $N^i dt$ and the spatial distance dx^i. The distance $N^i dt + dx^i$ is first measured by 3-metric h_{ij}, and then together with Ndt, makes ds.

The unit normal to the $t = \text{const}$ slices is $n_\mu \propto \partial_\mu t$; fixing $g^{\mu\nu} n_\mu n_\nu = -1$ (timelike, unit-normalized) sets the proportionality constant to N:

$$n_\mu = -N\,\partial_\mu t = (-N, 0, 0, 0), \qquad n^\mu = g^{\mu\nu} n_\nu = \left(\frac{1}{N}, -\frac{N^i}{N}\right). \tag{33}$$

(Check: $g^{\mu\nu} n_\mu n_\nu = g^{00} N^2 = -1$.) The induced metric is then the projector orthogonal to n^μ,

$$h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu, \tag{34}$$

which reduces to h_{ij} on spatial indices and satisfies $h_{\mu\nu} n^\nu = 0$ by construction.

Extrinsic curvature, $K_{\mu\nu} = h_\mu{}^\rho \nabla_\rho n_\nu$, measures how the normal tilts as you move along the slice — the "velocity" of the embedding. In ADM variables it has the closed form

$$K_{ij} = \frac{1}{2N}\left(\dot{h}_{ij} - D_i N_j - D_j N_i\right), \tag{35}$$

with D_i the covariant derivative built from h_{ij}. Read it the useful way around: this is just $\dot{h}_{ij} = 2NK_{ij} + D_i N_j + D_j N_i$ solved for K_{ij}, so K_{ij} *is* the time derivative of the induced metric, once the lapse/shift bookkeeping is stripped off.

The reframing that pays off: *these are precisely the objects covariant under the residual spatial diffeomorphisms.* ADM is not an imported tool — it is the direct consequence of the breaking pattern.

The menu of allowed building blocks then follows by inspection:

- $g^{00} = -1/N^2$: the one scalar covariant under spatial diffs but *not* under time diffs — and free functions $f(t)$ of the broken time. Hence $\delta g^{00} \equiv g^{00} + 1$ and coefficients $f(t)$ everywhere.
- $\delta K_{\mu\nu} = K_{\mu\nu} - a^2 H h_{\mu\nu}$: fluctuation of the extrinsic curvature.
- ${}^{(3)}R_{\mu\nu}$: intrinsic curvature of the slice (related to δK and bulk curvature by Gauss–Codazzi, not independent).
- Spatial covariant derivatives of all the above.

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2 Cosmology as Spontaneous Symmetry Breaking

In Lecture 1 we broke an internal $U(1)$; now we break *time translations.* The logic is identical, and we run it in the same order: diagnose the breaking through the *global* symmetry (here, a Killing vector), identify the Goldstone, and pass to unitary gauge. The one new subtlety is that "how broken" is a question with a quantitative answer — the order parameter turns out to be $\dot{H}$ — and de Sitter is the special point where the answer is "not broken."

2.1 When is a spacetime symmetry broken?

General relativity is a gauge theory: it is invariant under coordinate changes $x^\mu \to x'^\mu(x)$, with $g_{\mu\nu}$ the gauge field. As in Lecture 1, a gauge symmetry is a redundancy and cannot be "broken" literally — so we diagnose the breaking through its *global* version.

The global version of a time reparametrization $t \to t'(x)$ is a rigid time translation, and a continuous isometry is generated by a *Killing vector* ξ^μ,

$$\mathcal{L}_\xi g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = 0. \tag{36}$$

Time translations correspond to a *timelike* Killing vector, $\xi^2 \equiv g_{\mu\nu}\xi^\mu\xi^\nu < 0$. Hence the criterion:

Time translations are spontaneously broken by any solution whose metric admits *no* timelike Killing vector — i.e. by any non-stationary spacetime.

The vacuum here is the background *spacetime*, and the "order parameter" is whether it sits still in time. Let us verify different examples

- **Minkowski:** With $g_{\mu\nu} = \eta_{\mu\nu}$ constant, the vector $\xi = \partial_t$ (i.e. $\xi^\mu = \delta^\mu_0$) satisfies $\mathcal{L}_{\partial_t}\eta_{\mu\nu} = 0$ trivially, and $\xi^2 = \eta_{00} = -1 < 0$. Time translations are unbroken: Minkowski is static.
- **dS:** Inflation is close to de Sitter, so we must understand the de Sitter case first — and it is subtle. In flat slicing,

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + e^{2Ht}\,\mathrm{d}\mathbf{x}^2 \qquad (H = \text{const}). \tag{37}$$

  This is *not* static: ∂_t fails the Killing equation. Yet de Sitter is maximally symmetric (ten Killing vectors), and one of them combines a time translation with a compensating spatial *dilation*:

$$t \to t + \Delta t, \qquad \mathbf{x} \to e^{-H\Delta t}\,\mathbf{x}. \tag{38}$$

  The time shift makes $a = e^{Ht}$ grow; rescaling the comoving coordinates by $e^{-H\Delta t}$ undoes it exactly. Infinitesimally ($\Delta t = \epsilon$), the generator is

$$\xi = \partial_t - H\,x^i\partial_i, \tag{39}$$

  or in components ($\xi^0 = 1,\ \xi^i = -Hx^i$).

- **Is ξ Killing?** The 00 and $0i$ components of $\mathcal{L}_\xi g_{\mu\nu}$ vanish quickly. The spatial block is the instructive one:

$$\mathcal{L}_\xi g_{ij} = \underbrace{\xi^0\partial_0 g_{ij}}_{2a^2H\,\delta_{ij}} + \underbrace{g_{kj}\partial_i\xi^k + g_{ik}\partial_j\xi^k}_{-2a^2H\,\delta_{ij}} = 0, \tag{40}$$

  where the second group used $\partial_i\xi^k = -H\delta^k_i$ and $g_{ij} = a^2\delta_{ij}$. The growth of the scale factor ($+2a^2H$) is exactly cancelled by the dilation ($-2a^2H$). So ξ is a genuine Killing vector of de Sitter.

- **Is it timelike?** Compute its norm,

$$\xi^2 = g_{00}(\xi^0)^2 + g_{ij}\xi^i\xi^j = -1 + e^{2Ht}\,H^2|\mathbf{x}|^2. \tag{41}$$

  Thus ξ is timelike ($\xi^2 < 0$) precisely when

$$e^{Ht}|\mathbf{x}| < H^{-1}, \tag{42}$$

  i.e. when the *physical* distance from the origin, $a|\mathbf{x}| = e^{Ht}|\mathbf{x}|$, is smaller than the Hubble radius H^{-1}. The Killing vector is timelike everywhere inside the horizon, null on it, spacelike outside. For all local physics, then, de Sitter behaves as a state of gravity with *unbroken* time translations: the combined transformation is a symmetry, the "clock" does not tick.

- **q–dS and FLRW:** Inflation is *quasi*–de Sitter: H varies slowly rather than being exactly constant.

$$ds^2 = -dt^2 + a^2(t)\delta_{ij}dx^i dx^j \tag{43}$$

  Define the first slow-roll parameter

$$\epsilon \equiv -\frac{\dot{H}}{H^2}, \qquad \text{quasi-dS}: \quad 0 < \epsilon \ll 1, \qquad a(t) = \exp\int H\,\mathrm{d}t. \tag{44}$$

Exact de Sitter is the point $\epsilon = 0$. Now ask: does the dilation survive when $\dot{H} \neq 0$? Take the same candidate $\xi = \partial_t - H(t)x^i\partial_i$ on the general FLRW metric and compute $\mathcal{L}_\xi g_{\mu\nu}$ component by component:

$$\mathcal{L}_\xi g_{00} = 2g_{00}\,\partial_0\xi^0 = 0, \tag{45}$$

$$\mathcal{L}_\xi g_{ij} = 2a^2H\,\delta_{ij} - 2a^2H\,\delta_{ij} = 0 \qquad (\text{for } \textit{any } a(t)), \tag{46}$$

$$\mathcal{L}_\xi g_{0i} = g_{ki}\,\partial_0\xi^k = a^2\delta_{ki}\,\partial_0(-Hx^k) = -a^2\dot{H}\,x_i\,. \tag{47}$$

The 00 and ij blocks vanish identically — the dilation always handles the isotropic stretching. The *only* obstruction to the Killing equation is the time–space mixing, and it is proportional to $\dot{H}$:

$\dot{H}$ is the order parameter of the breaking. Exact dS ($\dot{H} = 0$): ξ is Killing, time translations unbroken. Quasi-dS ($\dot{H} \neq 0$): ξ fails at $\mathcal{O}(\dot{H})$, time translations genuinely broken.

Empirical check. Two facts say $\epsilon \neq 0$: inflation must end (the accelerating phase exits, so H cannot be exactly constant), and the primordial spectrum has a red tilt,

$$n_s - 1 \simeq -2\epsilon - \eta < 0, \qquad n_s \approx 0.965, \tag{48}$$

measured away from unity. The near-scale-invariance ($\epsilon \ll 1$) is why the background is close to de Sitter; the deviation ($\epsilon \neq 0$) is why there is a Goldstone at all. By the Lecture-1 logic, spontaneous breaking of time translations comes with a Goldstone $\pi(x)$, made explicit by the Stückelberg trick

$$t \to t + \pi(x). \tag{49}$$

Under the *unbroken* spatial translations and rotations, π transforms linearly — it is a three-dimensional scalar. Rather than *postulating* an inflaton scalar *ab initio*, the EFT shows a scalar is the *inevitable consequence* of broken time translations — the Goldstone that any non–de Sitter expansion must carry. It answers the unease about fundamental scalars in inflation: the scalar was never optional.

2.2 The most general action and tadpole cancellation

Assemble every operator from the Lecture-1 menu:

$$S = \int \mathrm{d}^4x\,\sqrt{-g}\left[\frac{M_{\rm pl}^2}{2}R + \underbrace{M_{\rm pl}^2\dot{H}g^{00} - M_{\rm pl}^2\big(3H^2+\dot{H}\big)}_{\text{background (fixed)}} + \sum_{n\geq 2}\frac{M_n^4(t)}{n!}\,(\delta g^{00})^n \right.$$
$$\left. - \frac{\bar{M}_1^3(t)}{2}\delta g^{00}\delta K - \frac{\bar{M}_2^2(t)}{2}\delta K^2 - \frac{\bar{M}_3^2(t)}{2}\delta K_\mu{}^\nu \delta K_\nu{}^\mu + \cdots\right]. \tag{50}$$

The action splits in two: a background piece and a fluctuation piece. *Tadpole cancellation* fixes the background. Requiring that the FLRW solution extremize the action — no terms linear in the fluctuations — forces the coefficients of the constant and of g^{00} to the values shown. Equivalently, they follow from the Friedmann equations. The background is *not* free: it is dictated by demanding that the chosen $H(t)$ solve the equations of motion.

> **Free coefficients:** $M_n^4(t)$ for $n \geq 2$ and $\bar{M}_i(t)$.
> Everything at the tadpole level is fixed by $H(t)$.

2.3 The universal part is slow-roll in disguise

Keep only the background piece:

$$S_0 = \int \mathrm{d}^4x\,\sqrt{-g}\left[\frac{M_{\rm pl}^2}{2}R - M_{\rm pl}^2\big(3H^2+\dot{H}\big) + M_{\rm pl}^2\dot{H}\,g^{00}\right]. \tag{51}$$

Insert a canonical scalar on its background, $\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 - V(\phi)$ with $\phi = \bar{\phi}(t)$, so $-\frac{1}{2}(\partial\phi)^2 \to -\frac{1}{2}\dot{\bar{\phi}}^2\,g^{00}$. Using $\dot{\bar{\phi}}^2 = -2M_{\rm pl}^2\dot{H}$ and $V = M_{\rm pl}^2(3H^2+\dot{H})$ reproduces S_0 exactly. *Every* single-clock model shares this spine; the M_n^4 encode the departures.

2.4 Stückelberg in gravity

Restore the broken time diffs by reintroducing π, $t \to t+\pi$. Under this,

$$g^{00} \to g^{\mu\nu}\partial_\mu(t+\pi)\partial_\nu(t+\pi) = g^{00} + 2\,g^{0\mu}\partial_\mu\pi + g^{\mu\nu}\partial_\mu\pi\,\partial_\nu\pi, \tag{52}$$

$$M_n^4(t) \to M_n^4(t+\pi), \qquad H(t) \to H(t+\pi), \quad \text{etc.} \tag{53}$$

This is the gauge toy model, verbatim, with time diffs in place of $U(1)$.

2.5 Decoupling limit, sound speed, dispersion

The mixing with metric perturbations switches off in the decoupling limit,

$$M_{\rm pl} \to \infty, \quad \dot{H} \to 0, \quad M_{\rm pl}^2\dot{H} \text{ fixed}, \tag{54}$$

so above $\omega^2 > \omega_{\rm mix}^2 = |\dot{H}|$ we evaluate π on the unperturbed FLRW background (the mirror of $E_{\rm mix} = m$). The Goldstone action becomes, schematically,

$$S_\pi = \int \mathrm{d}^4x\,\sqrt{-g}\left[M_{\rm pl}^2\dot{H}\,(\partial\pi)^2 + \sum_{n\geq 2}\frac{M_n^4}{n!}\left(-2\dot{\pi} + (\partial\pi)^2\right)^n\right]. \tag{55}$$

At quadratic order,

$$\mathcal{L}_\pi^{(2)} = M_{\rm pl}^2\dot{H}\,(\partial\pi)^2 + 2M_2^4\,\dot{\pi}^2 \quad \Longrightarrow \quad \boxed{c_s^2 = \frac{M_{\rm pl}^2\dot{H}}{M_{\rm pl}^2\dot{H} - 2M_2^4} \leq 1} \tag{56}$$

and the dispersion relation $\omega^2 = c_s^2k^2$. If instead the extrinsic-curvature operators $\bar{M}_i$ dominate the spatial gradients, the dispersion turns over to $\omega^2 \propto k^4$ — flag this now as the ghost-inflation teaser for Lecture 3.

3 Stückelberg Action in the decoupling limit

The fourth order action in unitary gauge is given by

$$
\begin{aligned}
S[\delta g^{00},\delta K^\mu_\nu] = \int d^4x\sqrt{-g}\Bigg[&\frac{M_{\rm pl}^2}{2}R + M_{\rm Pl}^2\dot H g^{00} - M_{\rm Pl}^2(3H^2+\dot H) + \frac{M_2^4(t)}{2!}(\delta g^{00})^2 + \frac{M_3^4(t)}{3!}(\delta g^{00})^3 \\
&+ \frac{M_4^4(t)}{4!}(\delta g^{00})^4 - \frac{\bar M_1^3(t)}{2}(\delta g^{00})\,\delta K^\mu{}_\mu - \frac{\bar M_2^2(t)}{2}(\delta K^\mu{}_\mu)^2 - \frac{\bar M_3^2(t)}{2}\delta K^\mu{}_\nu\delta K^\nu{}_\mu - \frac{\bar M_4^3(t)}{3!}(\delta g^{00})^2\delta K^\mu{}_\mu \\
&- \frac{\bar M_5^2(t)}{3!}(\delta g^{00})(\delta K^\mu{}_\mu)^2 - \frac{\bar M_6^2(t)}{3!}(\delta g^{00})\delta K^\mu{}_\nu\delta K^\nu{}_\mu - \frac{\bar M_7(t)}{3!}(\delta K^\mu{}_\mu)^3 - \frac{\bar M_8(t)}{3!}\delta K^\mu{}_\mu\delta K^\nu{}_\rho\delta K^\rho{}_\nu \\
&- \frac{\bar M_9^1(t)}{3!}\delta K^\mu{}_\nu\delta K^\nu{}_\rho\delta K^\rho{}_\mu - \frac{\bar M_{10}^3(t)}{4!}(\delta g^{00})^3\delta K^\mu{}_\mu - \frac{\bar M_{11}^2(t)}{4!}(\delta g^{00})^2(\delta K^\mu{}_\mu)^2 - \frac{\bar M_{12}^2(t)}{4!}(\delta g^{00})^2\delta K^\mu{}_\nu\delta K^\nu{}_\mu \\
&- \frac{\bar M_{13}^1(t)}{4!}(\delta g^{00})(\delta K^\mu{}_\mu)^3 - \frac{\bar M_{14}^1(t)}{4!}(\delta g^{00})\delta K^\mu{}_\mu\delta K^\nu{}_\rho\delta K^\rho{}_\nu - \frac{\bar M_{15}^1(t)}{4!}(\delta g^{00})\delta K^\mu{}_\nu\delta K^\nu{}_\rho\delta K^\rho{}_\mu \\
&- \frac{\bar M_{16}^0(t)}{4!}(\delta K^\mu{}_\mu)^4 - \frac{\bar M_{17}^0(t)}{4!}(\delta K^\mu{}_\mu)^2\delta K^\nu{}_\rho\delta K^\rho{}_\nu - \frac{\bar M_{18}^0(t)}{4!}\left(\delta K^\mu{}_\nu\delta K^\nu{}_\mu\right)^2 - \frac{\bar M_{19}^0(t)}{4!}\delta K^\mu{}_\mu\delta K^\nu{}_\rho\delta K^\rho{}_\sigma\delta K^\sigma{}_\nu \\
&- \frac{\bar M_{20}^0(t)}{4!}\delta K^\mu{}_\nu\delta K^\nu{}_\rho\delta K^\rho{}_\sigma\delta K^\sigma{}_\mu\Bigg].
\end{aligned}
\tag{57}
$$

The superscript on M, $\bar M$, denotes the mass dimension of the coefficient. It seems that combinations of these operators which are odd under time reversal, $t\to -t$, can be removed by a disformal transformation. Although such a transformation is not a symmetry of the action, it leaves invariant correlation functions.

Restoring the Goldstone through the broken time reparameterization

$$t \to t + \pi(x), \tag{58}$$

in the decoupling limit yields,

$$\delta g^{00} \to -2\dot\pi - \dot\pi^2 + \frac{(\partial_i\pi)^2}{a^2(t)} = -2\dot\pi + (\partial_\mu\pi)^2, \tag{59}$$

$$
\begin{aligned}
K_{ij} \to \frac{1}{\sqrt{1-\delta g^{00}}}\Bigg[&- a^2H(1+\dot\pi)\delta_{ij} + \partial_{ij}\pi - \frac{(1+\dot\pi)}{1-\delta g^{00}}\bigg(2\partial_{(i}\dot\pi\partial_{j)}\pi - \frac{(1+\dot\pi)}{1-\delta g^{00}}\ddot\pi\partial_i\pi\partial_j\pi \\
&+ \frac{\partial_i\pi\partial_j\pi\partial^k\pi\,(2\partial_k\dot\pi - H\partial_k\pi)}{a^2(1-\delta g^{00})}\bigg) + \frac{2\partial^k\pi\partial_{(i}\partial_k\pi\partial_{j)}\pi}{a^2(1-\delta g^{00})} + \frac{\partial_i\pi\partial_j\pi\partial^k\pi\partial^l\pi\partial_{kl}\pi}{a^4(1-\delta g^{00})^2}\Bigg].
\end{aligned}
\tag{60}
$$

Unlike in unitary gauge, the K_{00} and K_{0i} components are non-vanishing and proportional to π. Since $n^\nu K_{\mu\nu} = 0$, these components are completely determined by K_{ij} as,

$$K_{00} = -\frac{n^j}{n^0}K_{0j}, \qquad K_{0j} = -\frac{n^i}{n^0}K_{ij}, \qquad \text{with} \quad \frac{n^i}{n^0} = -\frac{\delta^{ij}\partial_j\pi}{(1+\dot\pi)a^2}, \tag{61}$$

Notice that the relevant expressions entering the action are δK^μ_μ, $\delta K^\nu_\mu\delta K^\mu_\nu$, ..., etc., but thanks to Eq. (61) one finds the following:

$$K_\mu{}^\mu = \left(g^{ij} - \frac{n^in^j}{n_0^2}\right)K_{ij} \equiv \tilde h^{ij}K_{ij} = K_i{}^i, \qquad K_\mu{}^\nu K_\nu{}^\mu = K_i{}^jK_j{}^i, \tag{62}$$

where $\tilde h^{kj}h_{ik} = \delta_i^j$. We are interested in writing the fourth order action for π, thus $K_i{}^j$ up to cubic order reads

$$
\begin{aligned}
K_i{}^j = H\delta_i^j &- (1-\dot\pi)\frac{\partial^j\partial_i\pi}{a^2} + \frac{\partial_i\dot\pi\partial^j\pi}{a^2} + \frac{\partial_i\pi\partial^j\dot\pi}{a^2} - \frac{H}{a^2}\left(\partial_i\pi\partial^j\pi - \frac{1}{2H}\delta_i^j(\partial_k\pi)^2\right) \\
&+ \frac{\partial_i\pi\partial^j\pi}{a^2}(2H\dot\pi - \ddot\pi) - \frac{2\dot\pi}{a^2}\left(\partial_i\dot\pi\partial^j\pi + \partial_i\pi\partial^j\dot\pi\right) - \frac{\partial^j\partial_i\pi}{a^2}\left(\dot\pi^2 + \frac{(\partial_k\pi)^2}{2a^2}\right) - \frac{\delta_i{}^jH\dot\pi(\partial_k\pi)^2}{a^2} - \frac{\partial_i\pi\partial_k\pi\partial^{kj}\pi}{a^4}
\end{aligned}
\tag{63}
$$

Now we can write the action for π order by order. As usual, we will consider the Wilson coefficients as constants. Let us introduce

$$\boxed{f_\pi^4 \equiv 2\left(M_{\rm pl}^2|\dot H| + 2M_2^4\right), \qquad c_s^2 \equiv \frac{M_{\rm pl}^2|\dot H|}{M_{\rm pl}^2|\dot H| + 2M_2^4}, \qquad \tilde c_s^2 \equiv c_s^2 + \frac{\bar M_1^3H}{f_\pi^4}, \qquad \Lambda \equiv f_\pi\tilde c_s.} \tag{64}$$

Additionally, we can introduce the canonical field $\pi_c \equiv f_\pi^2 \pi$, so that the quadratic action in conformal time reads

$$\boxed{S^{(2)}_{\pi_c} = \int d\eta\, d^3x\; a^2(\eta) \left[\frac{{\pi_c'}^2}{2} - \tilde{c}_s^2 \frac{(\partial_i \pi_c)^2}{2} - \frac{\bar{c}_2}{\Lambda^2} \frac{(\nabla^2 \pi_c)^2}{a^2(\eta)} \right],} \tag{65}$$

where we have introduced the dimensionless coefficient $\bar{c}_2$ as $\bar{M}_2^2 + \bar{M}_3^2 = 2\,\bar{c}_2\; f_\pi^4/\Lambda^2$.

$$\boxed{\begin{aligned} S^{(3)}_{\pi_c} = \int d\eta\, d^3x\, a^4(\eta) \Bigg[& \frac{\mathcal{C}_1}{\Lambda^2} \frac{{\pi_c'}^3}{a^3} + \frac{\mathcal{C}_2}{\Lambda^2} \frac{\pi_c'(\partial_i \pi_c)^2}{a^3} + \frac{\mathcal{C}_3}{\Lambda^3} \frac{\dot{\pi}_c^2 \nabla^2 \pi_c}{a^4} + \frac{\mathcal{C}_4}{\Lambda^4} \frac{\partial_i \pi_c \partial^i \pi_c' \nabla^2 \pi_c}{a^5} + \frac{\mathcal{C}_5}{\Lambda^4} \frac{\pi_c'(\partial_{ij} \pi_c)^2}{a^5} \\ & + \frac{\mathcal{C}_6}{\Lambda^4} \frac{\pi_c'(\nabla^2 \pi_c)^2}{a^5} + \frac{\mathcal{C}_7}{\Lambda^3} \frac{(\partial_i \pi_c)^2 \nabla^2 \pi_c}{a^4} + \frac{\mathcal{C}_8}{\Lambda^5} \frac{(\nabla^2 \pi_c)^3}{a^6} + \frac{\mathcal{C}_9}{\Lambda^5} \frac{(\nabla^2 \pi_c)(\partial_{ij} \pi_c)^2}{a^6} + \frac{\mathcal{C}_{10}}{\Lambda^5} \frac{\partial_{ik} \pi_c \partial_{ij} \pi_c \partial_{jk} \pi_c}{a^6} \Bigg]. \end{aligned}} \tag{66}$$

$$\boxed{\begin{aligned} S^{(4)}_{\pi_c} = \int d\eta\, d^3x\, a^4(\eta) \Bigg[& \frac{\mathcal{D}_1}{\Lambda^4} \frac{{\pi_c'}^4}{a^4} + \frac{\mathcal{D}_2}{\Lambda^4} \frac{{\pi_c'}^2 (\partial_i \pi_c)^2}{a^4} + \frac{\mathcal{D}_3}{\Lambda^5} \frac{{\pi_c'}^3 \partial^2 \pi_c}{a^5} + \frac{\mathcal{D}_4}{\Lambda^4} \frac{(\partial_i \pi_c)^4}{a^4} + \frac{\mathcal{D}_5}{\Lambda^6} \frac{\partial_i \pi_c \partial^i \pi_c' \partial_j \pi_c \partial^j \pi_c'}{a^6} \\ & + \frac{\mathcal{D}_6}{\Lambda^6} \frac{(\partial_i \pi_c)^2 (\partial_j \pi_c')^2}{a^6} + \frac{\mathcal{D}_7}{\Lambda^5} \frac{\pi_c' \partial_i \pi_c \partial_j \pi_c \partial^{ij} \pi_c}{a^5} + \frac{\mathcal{D}_8}{\Lambda^6} \frac{\pi_c' \partial_i \pi_c \partial_j \pi_c' \partial^{ij} \pi_c}{a^6} + \frac{\mathcal{D}_9}{\Lambda^6} \frac{\pi_c' \partial_i \pi_c \partial^i \pi_c' \partial^2 \pi_c}{a^6} \\ & + \frac{\mathcal{D}_{10}}{\Lambda^6} \frac{{\pi_c'}^2 (\partial^2 \pi_c)^2}{a^6} + \frac{\mathcal{D}_{11}}{\Lambda^7} \frac{\partial^i \pi_c \partial_{ij} \pi_c \partial^{jk} \pi_c \partial_k \pi_c'}{a^7} + \frac{\mathcal{D}_{12}}{\Lambda^7} \frac{\pi_c' \partial_j \partial^k \pi_c \partial^{ij} \pi_c \partial_{ik} \pi_c}{a^7} + \frac{\mathcal{D}_{13}}{\Lambda^6} \frac{\partial_i \pi_c \partial^j \pi_c \partial_{jk} \pi_c \partial^{ik} \pi_c}{a^6} \\ & + \frac{\mathcal{D}_{14}}{\Lambda^7} \frac{\partial_i \pi_c \partial^i \pi_c' (\partial_{jk} \pi_c)^2}{a^7} + \frac{\mathcal{D}_{15}}{\Lambda^6} \frac{(\partial_i \pi_c)^2 (\partial_{jk} \pi_c)^2}{a^6} + \frac{\mathcal{D}_{16}}{\Lambda^6} \frac{\partial^i \pi_c \partial^j \pi_c \partial_{ij} \pi_c\, \partial^2 \pi_c}{a^6} + \frac{\mathcal{D}_{17}}{\Lambda^7} \frac{\partial^i \pi_c \partial^j \pi_c' \partial_{ij} \pi_c\, \partial^2 \pi_c}{a^7} \\ & + \frac{\mathcal{D}_{18}}{\Lambda^7} \frac{\pi_c' (\partial_{ij} \pi_c)^2\, \partial^2 \pi_c}{a^7} + \frac{\mathcal{D}_{19}}{\Lambda^8} \frac{(\partial^2 \pi_c)^4}{a^8} + \frac{\mathcal{D}_{20}}{\Lambda^8} \frac{(\partial_{ij} \pi_c)^2 (\partial^2 \pi_c)^2}{a^8} + \frac{\mathcal{D}_{21}}{\Lambda^8} \frac{(\partial_{ij} \pi_c)^4}{a^8} + \frac{\mathcal{D}_{22}}{\Lambda^8} \frac{\partial_j \partial^k \pi_c \partial^{ij} \pi_C \partial_{ik} \pi_c\, \partial^2 \pi_c}{a^8} \Bigg]. \end{aligned}} \tag{67}$$

4 Model Space, Observables, and Dark Energy

The tightest lecture. Decide the observables/DE trade-off before you walk in (see closing note).

4.1 The model map: corners of $\{M_n^4\}$

Recall the π action, then read off three corners of coefficient space. These are *sub-cases* of the EFT, not UV completions: a UV completion lives *above* the cutoff, while these are choices of coefficients *inside* it.

Model	Coefficient choice	Signature
Canonical slow-roll	$c_s = 1$; all M_n^4 $(n \geq 2)$ off	Minimal case; universal part only
$P(X)$ / DBI	M_2^4 on $\Rightarrow$ $c_s < 1$; DBI = *specific relation* among the M_n^4 from $\sqrt{1-X}$	Fixed $\dot{\pi}^3$ coefficient $\Rightarrow$ a concrete non-Gaussianity distinct from generic $P(X)$
Ghost inflation	$\dot{H} \to 0$: kills $M_{\rm pl}^2 \dot{H}(\partial\pi)^2$; extrinsic-curvature ops take over	$\omega^2 \propto k^4$; callback to the Lecture-1 building blocks

Ghost inflation is the case to feature: it is where the extrinsic-curvature operators — decoration in slow-roll and DBI — become the *entire* dynamics. It justifies, after the fact, the trouble taken in Lecture 1.

4.2 Observables

Power spectrum. With $\zeta = -H\pi$,

$$\Delta_\zeta^2 = \frac{1}{4\pi^2}\left(\frac{H}{f_\pi}\right)^4, \qquad f_\pi^4 = 2M_{\rm pl}^2|\dot{H}|\,c_s. \tag{68}$$

Scale invariance is exact in de Sitter. The measured $\Delta_\zeta^2 \approx 2\times10^{-9}$ sets $f_\pi \approx 58\,H$.

Non-Gaussianity. A small c_s amplifies interactions. The leading single-derivative cubic terms are

$$\mathcal{L}_\pi^{(3)} \supset -\frac{1}{2\Lambda^2}\left[\dot{\pi}_c\frac{(\partial_i\pi_c)^2}{a^2} + A\,\dot{\pi}_c^3\right], \qquad A \simeq \left(-1+\tfrac{2}{3}\frac{M_3^4}{M_2^4}\right)c_s^2, \tag{69}$$

giving

$$f_{\rm NL} \sim \left(\frac{f_\pi}{\Lambda}\right)^2 \sim \frac{1}{c_s^2}, \qquad \text{equilateral shape.} \tag{70}$$

The coefficient A (through M_3^4) is what separates DBI from a generic $P(X)$. Close with the observational chain and the hierarchy of scales:

$$\pi \;\to\; \zeta \;\to\; \delta T \;\to\; \delta g. \tag{71}$$

4.3 Extension to dark energy

Same construction, new regime. Now the clock is dark energy, the background is a general FLRW with $w(t) \neq -1$, and *matter is present.* Two consequences reshape the treatment:

1. The Goldstone π is the dark-energy perturbation, no longer directly ζ.
2. No clean decoupling limit. At late times gravity is dynamical and we care how dark energy modifies it (effective Newton constant, gravitational slip), so we *keep* the metric. The operators subdominant during inflation — δK, ${}^{(3)}R$, the $\bar{M}_i$ — now carry the physics. This is the payoff of having built them in Lecture 1.

The action generalizes to (Gubitosi, Piazza & Vernizzi, arXiv:1210.0201; allow a time-dependent Planck mass through $f(t)R$):

$$\begin{aligned} S = \int \mathrm{d}^4x\,\sqrt{-g}\Big[&\frac{M_{\rm pl}^2}{2}f(t)R - \Lambda(t) - c(t)g^{00} + \frac{M_2^4(t)}{2}(\delta g^{00})^2 - \frac{\bar{M}_1^3(t)}{2}\delta g^{00}\,\delta K \\ &- \frac{\bar{M}_2^2(t)}{2}\delta K^2 - \frac{\bar{M}_3^2(t)}{2}\delta K_{\mu\nu}\delta K^{\mu\nu} + \cdots\Big]. \end{aligned} \tag{72}$$

Repackage the free functions into the dimensionless α_i basis (Bellini & Sawicki, arXiv:1404.3713), the natural endpoint:

Parameter	Name	Role / source operator
α_K	kineticity	scalar kinetic energy; from M_2^4
α_B	braiding	metric–scalar kinetic mixing; from $\bar{M}_1^3$
$\alpha_M = \dfrac{\mathrm{d}\ln M_*^2}{\mathrm{d}\ln a}$	Planck-mass run rate	from $\dot{f}(t)$
$\alpha_T = c_T^2 - 1$	tensor speed excess	from $\bar{M}_2^2, \bar{M}_3^2$

Horndeski connection. The dictionary from Horndeski/Galileon functions to $\{\alpha_i\}$ exists — show that it exists, do not grind through it.

The strong closing. α_T sets the gravitational-wave speed. GW170817 fixed $|c_T - 1| \lesssim 10^{-15}$, hence $\alpha_T \approx 0$, eliminating broad classes of Horndeski at a stroke. End on the spine:

One construction — broken time diffeomorphisms, a preferred foliation, the same building blocks. Two regimes.

Appendix A — Convention flags to resolve

- Fix all signs against Piazza–Vernizzi §2: the tadpole coefficients, the sign inside $(-2\dot{\pi} + (\partial\pi)^2)^n$, and the $\bar{M}_i$ normalizations.
- Metric signature: $(-, +, +, +)$ throughout (as above).
- $M_{\rm pl}^2 = 1/(8\pi G)$; $\dot{H} < 0$ during inflation, so $c_s^2 \leq 1$ needs $M_2^4 > 0$.
- $\delta K_{\mu\nu}$ background subtraction: $K_{\mu\nu} = a^2 H h_{\mu\nu}$ on FLRW; verify the a-powers in your slicing.

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